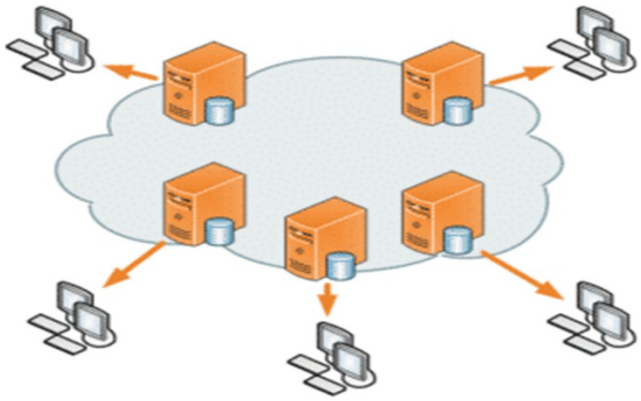
DCS SHEET#1 FALL 2026

1. Define Distributed Computer System DCS? What are the pros and cons of Distributed system over centralized system?

A:

Define:



- A DCS is one in which **components** located at networked computers communicate and **coordinate** their actions only by-passing **messages**.
- It requires autonomous computers to be interconnected through a network.
- Each computer has to be equipped with DOS software, which enables the computers to coordinate activities and to share resources in a controlled way
- DCS requires transparency of distribution for computer users. Users shall not be aware of the fact that the system is distributed

Pros over centralized system:

- **Scalability:**
  One of the primary advantages of DCS is scalability.
  As data volumes grow, organizations can expand their processing capabilities by adding more nodes to the network.
  This scalability ensures that the system can handle increasing workloads without a significant drop in performance, providing a flexible and adaptive solution to the challenges posed by big data.
- **Fault Tolerance:**
  DCSs inherently offer improved fault tolerance compared to centralized systems.
  In a distributed environment, if one node fails, the remaining nodes can continue processing data, reducing the risk of a complete system failure.
  This resilience is crucial for maintaining uninterrupted data operations in mission-critical applications.
- **Performance:**
  Parallel processing, a key feature of DCS, contributes to enhanced performance.
  By breaking down complex tasks into smaller subtasks distributed across nets, the system can process data more quickly and efficiently.
  This results in reduced processing times and improved overall performance, enabling organizations to derive insights from data in a timely manner
- **Efficient Handling of Large Volumes of Data:**
  In the era of big data, efficiently handling large volumes of data is a paramount concern for organizations.
  DCS cope with this aspect by employing data partitioning strategies.
  Large data sets are divided into smaller, more manageable segments, and each segment is processed independently across distributed nodes.
  This approach not only accelerates data processing through parallelism and clustering but also optimizes use of resources. Each node focuses on a specific subset of the data, ensuring that the system operates efficiently and effectively.
  The ability to efficiently handle large volumes of data positions organizations to extract meaningful insights, identify patterns, and make informed decisions.

2. Define data centers and mention its applications?

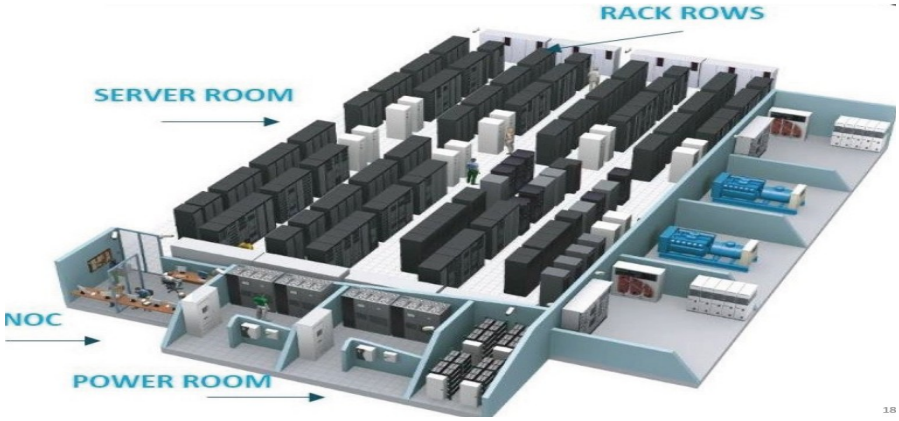
A:

- Data Centers DCs are the physical facilities that house the IT equipment and infrastructure necessary to store, process, and distribute vast and ever-growing volume of data.
- The housed and supported IT equipment including: servers, storage systems, network devices, cybersecurity devices, and other infrastructure components.
- DCs provide the physical infrastructure and environmental conditions necessary for the secured, reliable and efficient operation of these IT systems.
- DCs are typically designed with multiple levels of redundancy to ensure that they can continue to operate even in the event of a power outage, hardware failure, or other disruption.
- They employ advanced security physical and logical measures to protect the data and equipment from unauthorized access and cyber threats

3. Mention Data Centers components?

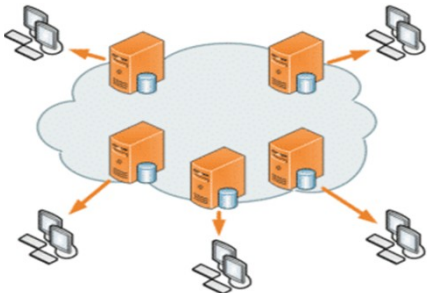
A:

- Human Resources
- Physical Suitable Site
- Racks and enclosures
- Power Distribution Units PDUs
- Cooling systems
- Fire suppression systems
- Security systems
- Computer resources
- Storage systems
- Network equipment
- Software and virtualization
- Data Center Management



4. Explain the key components of distributed computing systems using drawing?

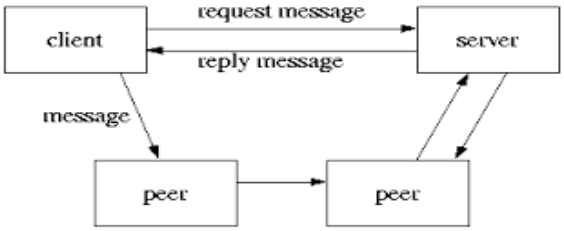
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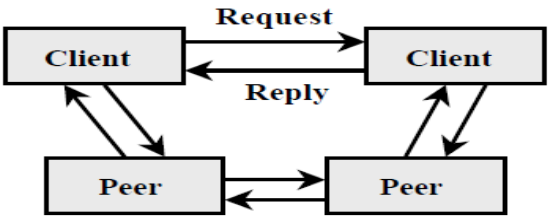


- **Geographical Distribution**:
  - Nodes are spread across different physical locations.
  - They communicate via a network, such as a local area network (LAN) or the internet.
- **Resource Sharing**:
  - Nodes share resources such as processing power, storage, and data.
  - This enables more efficient utilization of resources.
- **Concurrency**:
  - Multiple nodes operate concurrently, performing tasks simultaneously.
  - This parallelism enhances the system's overall performance and throughput.
- **Scalability**:
  - Easy to scale by adding more nodes to the system.
  - System capacity and performance improve with the addition of resources.
- **Fault Tolerance** :
  - Designed to handle failures gracefully.
  - Redundancy and replication ensure the system remains operational even if some nodes fail.
- **Transparency**:
  - The complexity of the distributed system is hidden from users.
  - Users interact with the system as if it were a single entity.

5. With the help of a diagram explain the working of distributed systems.

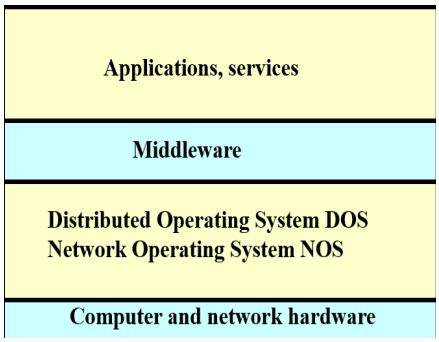
A:





6. Draw the DCS software architecture and compare between distributed operating system and network operating system?

A:



A NOS is an operating system that is expected to support networking functionalities.

This enables to connect several computers and to share many resources, e.g. files, applications, and printers.

Although each computer works self-contained it can communicate with other users in the network through the NOS.

NOS Examples: MS SERVER, IBM AIX, ORCLE SOLARES, LINUX SERVER, UNIX

- Advantages of Network OS:
- Resource Sharing
- Centralized Control
- Scalability
- NOS Security

DOS is responsible for coordinating the computers/nodes that are connected and operates in cooperation like a single large machine.

In DOS every computer operates as an individual OS, a DOS unifies all computers and they work together.

DOS EXAMPLES: MS AZURE, GOOGLE FILE SYSTEM GFS, LOCUS, ORCLE SOLARES

Advantages of Distributed OS:

- Resource Utilization
- Transparency
- Fault Tolerance
- Scalability
- DOS Security

7. Compare between centralized, decentralized, and distributed computing systems?

A:

ASPECT	CENTRALIZED SYSTEMS	DECENTRALIZED SYSTEMS	DISTRIBUTED SYSTEMS
DEFINITION	Single central server controls and manages all operations	Multiple nodes with independent control, no central authority.	Multiple interconnected nodes working together as a single system.
CONTROL	Centralized control with a single point of management	Distributed control, each node operates independently.	Shared control, nodes collaborate to achieve common goals.
SINGLE POINT OF FAILURE	High risk; if the central server fails, the whole system fails.	Reduced risk; failure of one node does not impact the entire system.	Reduced risk; designed for fault tolerance and redundancy.
SCALABILITY	Limited scalability, can become a bottleneck	More scalable, can add nodes independently.	Highly scalable, can add more nodes to distribute the load.
RESOURCE UTILIZATION	Central server resources are heavily utilized	Resources are spread across multiple nodes.	Efficient resource sharing across nodes.
PERFORMANCE	Can be high initially but may degrade with increased load	Generally good, performance improves with more nodes.	High performance due to parallel processing and resource sharing.
MANAGEMENT	Easier to manage centrally.	More complex, requires managing multiple nodes.	Complex, requires coordination and management of many nodes.
LATENCY	Lower latency, as operations are managed centrally.	Can vary, depends on the distance between nodes.	Potentially higher latency due to network communication.

8. Identify the various challenges in the distributed system.
A:

9. Explain what is meant by (distribution) transparency, and give examples of different types of transparency.
A:

- Transparency ensures that the DCS should be perceived as a single entity by the users or the application programmers rather than a collection of autonomous systems, which is cooperating.
- The user should be unaware of where the services are located and the transfer from a local machine to a remote one should be transparent

TRANSPARENCY	DESCRIPTION
Access	• Hide differences in data representation and how a resource is accessed
Location	• Hide where a resource is located, users cannot tell where hardware and software resources such as Servers, printers, files, data bases are located.
Migration	• Hide that a resource may move to another location
Relocation	• Hide that a resource may be moved to another location while in use
Replication	• Hide that a resource is replicated, OS can make additional copies of files and resources without users noticing.
Concurrency	• Hide that a resource may be shared by several competitive users, The users are not aware of the existence of other users. Need to allow multiple users to concurrently access the same resource. Lock and unlock for mutual exclusion.
Failure	• Hide the failure and recovery of a resource
Concurrency Transparency	• Automatic use of parallelism without having to program explicitly. The objectives of distributed and parallel system designers

10. Explain the differences between homogeneous and heterogenous systems?
A:

11. List the functions of middleware.
A:

12. Give examples for security threats and security protection.
A:

13. What is meant by scalability challenge and how to achieve it?
A:

14. What is meant by resources sharing challenge and how to achieve it?
A:

15. What is meant by openness challenge and how to achieve it?
A:

16. What is meant by fault tolerance challenge and how to achieve it?

A:

17. Explain and List the functions of middleware.
A:

18. Explain and List the functions of clustering computing.
A:

19. Explain and List the functions of virtualization.
A:

20. What are the benefits of a computer cluster?
A:

21. What are the types of cluster computing?
A:

GOOD LUCK